The Algebraic Degradation of the BITSian: A Study in Representation Theory, Dynkin Diagrams, and Spontaneous Symmetry Breaking[1]

Jay Agarwal
BITS Pilani
8076527607
f20221305@pilani.bits-pilani.ac.in

Abstract. "Taking it lite" the practice of doing nothing all semester while maintaining absolute confidence that the compre will sort itself out is the defining feature of student life at BITS Pilani. In this paper, we model the BITSian student using the representation theory of $U(1)$, the Lie algebra $\mathfrak{su}(5)$, and the Dynkin diagram of $SU(5)$, proving that academic and social life cannot simultaneously be maximised. We then apply spontaneous symmetry breaking to show that the lazy vacuum is not chosen, it is energetically preferred and that the AUGSD functions as an explicit symmetry-breaking term that gives mass to the procrastination Goldstone boson. The mathematics does not judge. It simply proves.

1. Introduction

Every semester, a new batch of students arrives at BITS Pilani with hope, ambition, and a firm plan to maintain above an 8 CG. The author had first written 9 CG but who are we kidding? By week three, most of them are at the ANC at 2 AM convincing themselves that the compre is "still far.", getting into one club, department and sports team is easy, they are not cringe and other lies we tell ourselves to maintain our delusion. The author is totally not leaking their first year thoughts here.

This paper makes no moral argument. We observe that the algebraic structure of the situation already contains every conclusion one might reach, and that the student, had they attended the relevant lectures, might have seen this coming. We model the student's state using the representation theory of compact Lie groups, classify their available life choices via the Dynkin diagram of $SU(5)$ and explain the universality of the lazy equilibrium using spontaneous symmetry breaking. The rationale for choosing this topic is that the authors are personally affected by the phenomena described herein.

2. The Lite Transformation: Representation Theory of $U(1)$

Model the student's academic potential as a complex number $\psi \in \mathbb{C}$. The group $U(1) = \{e^{i\theta} : \theta \in [0, 2\pi)\}$ acts on $\mathbb{C}$ by multiplication. Each application of $e^{i\theta}$ is a *Lite Transformation* the student rotates their worldview and convinces themselves a 6 CG is actually fine.

[1] The author would like to clarify that this paper was written entirely by a human being with full cognitive autonomy, and not at all assisted by any large language model. The author would also like to clarify that they are not just saying this. They are very much saying this. As we enter an era where AI threatens to displace every intellectual endeavour, it is important to note that the insights in this paper are the product of genuine human suffering, not matrix multiplications. Probably.

The irreducible representations of $U(1)$ are all one-dimensional, labelled by integers $n \in \mathbb{Z}$:

$$\rho_n : e^{i\theta} \mapsto e^{in\theta}, \qquad n \in \mathbb{Z}.$$

The student's state lives in ρ_1. By the Peter-Weyl theorem, any $U(1)$-representation decomposes into a direct sum of these irreps. The component of the student's academic life in ρ_0, the trivial representation is the part invariant under every Lite Transformation. This is your suffering. It transforms trivially. It is unaffected by how lite you take it.

The character of ρ_1 is:

$$\chi_1(e^{i\theta}) = e^{i\theta}.$$

As θ varies over $[0, 2\pi)$, this traces out the unit circle $S^1 \subset \mathbb{C}$.

The orbit of your potential under the Lite Transformation is a circle. You are going in circles. Not metaphorically, the character of ρ_1 computed your life and returned S^1.

Remark. *The student whose potential is in ρ_0 is fixed by everything: $e^{i\theta} \cdot \psi = \psi$ for all θ. He has achieved the trivial representation. He has, in the technical sense, maximal symmetry. We do not recommend this.*

3. Life Is Zero-Sum: The $\mathfrak{su}(5)$ Lie Algebra

A common objection: "I'm not taking academics lite, I'm just prioritising other things." (Sure)

Model the student as a vector in $\mathbb{C}^5$, the standard representation of $SU(5)$:

$$\Psi = \begin{pmatrix} \text{CGPA} \\ \text{Placements} \\ \text{Sleep} \\ \text{Physical Health} \\ \text{Social Life} \end{pmatrix}$$

(Note how the author hasn't included mental health here[2])

Since every element of $SU(5)$ is unitary, the action preserves $\|\Psi\|^2$. Total life-energy is conserved. You are not generating more of it.

The Lie algebra $\mathfrak{su}(5)$ consists of traceless skew-Hermitian 5×5 matrices. Define the *Lite Generator*:

$$H = i \cdot \mathrm{diag}(-4, -1, 1, 2, 2) \in \mathfrak{su}(5), \qquad \mathrm{Tr}(H) = 0.$$

Exponentiating via the exponential map $\exp : \mathfrak{su}(5) \to SU(5)$:

$$\gamma(t) = e^{tH} = \mathrm{diag}(e^{-4t}, e^{-t}, e^{t}, e^{2t}, e^{2t}).$$

[2]The attentive reader may notice that Ψ has five components and that certain significant aspects of college life have been omitted. Mental health has already been acknowledged. There is one further omission the author is aware of, and the reader, having lived in a BITS hostel past 11 PM, is certainly aware of too. It has not been included for reasons that are self-evident. Including it would require upgrading to $SU(6)$, changing the Dynkin diagram from A_4 to A_5, adding a node, and complicating every subsequent calculation in this paper. The author has therefore elected to omit it entirely in the interest of mathematical cleanliness. This is the only reason.

This is a smooth homomorphism $\gamma : \mathbb{R} \to SU(5)$ with $\gamma(0) = I$ and $\gamma(s+t) = \gamma(s)\gamma(t)$.

As t increases, CGPA decays as e^{-4t} while Social Life grows as e^{2t}. Their ratio is e^{6t}, growing exponentially without bound. You are not hacking the system. You are applying a perfectly valid one-parameter subgroup of $SU(5)$ that rotates your future into your present.

Remark. *The homomorphism γ has trivial kernel. Every moment of taking it lite is a distinct, non-repeatable element of the group. There is no inverse e^{-tH} button on the ERP.*

4. The Dynkin Diagram of Your Life

The Lie algebra $\mathfrak{su}(5)$ has rank 4, corresponding to the simple Lie algebra A_4, whose Dynkin diagram is:

The nodes are the simple roots of A_4. Two nodes are connected by an edge if and only if the corresponding roots are *not orthogonal*, they interfere with each other. Disconnected nodes represent choices that are mutually compatible.

Observe: CGPA and Social Life are the two endpoints of the diagram. They are not directly connected. Between them sit Placements and Sleep, mediating the conflict. This is not a coincidence. It is the Cartan matrix of A_4.

Theorem 4.1. *A Dynkin diagram of type A_n contains no cycles. CGPA and Social Life cannot be simultaneously connected to each other and to the intermediate nodes without creating a cycle, which would make the diagram affine ($\tilde{A}_4$) rather than finite-type.*

The student who believes they can max both CGPA and Social Life without sacrificing anything in between is implicitly claiming the Dynkin diagram of $SU(5)$ contains a cycle. By the classification of simple Lie algebras, this would make the associated algebra infinite-dimensional. The student is not wrong because they lack discipline. They are wrong because their life violates the Cartan–Killing classification.

Remark. *The dual-degree student at BITS navigates both A_4 (Physics) and A_4 (CS) simultaneously. Their combined root system has rank 8. The only rank-8 exceptional Lie algebra is E_8, which has dimension 248. The dual-degree student is managing 248 degrees of freedom. Their suffering is exceptional, in the technical sense. This has nothing to do with the fact that the author themselves are a dualite.*

5. The Lazy Vacuum: Spontaneous Symmetry Breaking

We now explain why taking it lite is not a choice. It is a ground state.

Model the student's "academic responsibility" as a complex scalar field $\phi \in \mathbb{C}$, where $|\phi|$ measures their current level of effort. The potential governing ϕ is:

$$V(\phi) = -\mu^2|\phi|^2 + \lambda|\phi|^4, \qquad \mu^2, \lambda > 0.$$

This is the standard Mexican hat potential. The point $\phi = 0$ is a local *maximum*, it is an unstable equilibrium. The student who attempts to remain in the "responsible" state is balancing at the top of the hat. The actual minima occur at:

$$|\phi_0| = \sqrt{\frac{\mu^2}{2\lambda}},$$

forming a degenerate circle of ground states. Every point on this circle is an equally valid minimum. They correspond to every equally bad way of being unproductive: watching anime, scrolling through reels, initiating conversations about which F1 driver would survive a zombie apocalypse. (The author has participated in all three, sometimes simultaneously, and can confirm the minimum is stable.)

The $U(1)$ symmetry $\phi \mapsto e^{i\alpha}\phi$ rotates among these minima. The student spontaneously breaks this symmetry by falling into one of them. This is not laziness. This is the system seeking its ground state. The author reached their ground state around Week 2 of Semester 1 and has been there since.

Theorem 5.1 (Goldstone)**.** *If a continuous symmetry is spontaneously broken, the theory contains a massless Goldstone boson for each broken generator.*

The broken generator here is the $U(1)$ rotation among lazy states. The corresponding Goldstone boson is *procrastination*, it costs zero energy to slide from one lazy activity to another along the circle of minima. Switching from playing Valorant to rewatching the same three episodes of a show you have already seen is a massless excitation. The restoring force is zero. Goldstone's theorem told you this. (The author discovered this empirically, somewhat before discovering Goldstone's theorem, and significantly before attending the relevant lecture.[3])

AUGSD as the Explicit Symmetry Breaking Term

If the story ended here, the student could coast indefinitely, sliding freely along the degenerate vacuum. The administration prevents this by introducing an explicit symmetry-breaking term into the potential:

$$V(\phi) \to V(\phi) - \varepsilon(\phi + \phi^*), \qquad \varepsilon > 0.$$

This term, generated by surprise quizzes, attendance weightages, and compre schedules, tilts the Mexican hat. The degenerate circle of minima is lifted: there is now one true minimum, and it is not the lazy one. The Goldstone boson acquires a mass:

$$m^2_{\text{procrastination}} \sim \varepsilon.$$

What this means physically: procrastination is no longer free. Every hour spent in the wrong minimum now carries a restoring force proportional to ε, pulling you back toward the exam you have not studied for. The AUGSD has not disciplined you. They have added a symmetry-breaking term to your effective potential and allowed the mass gap to do the work.

The size of ε is set by the administration. This is why the 0% attendance policy feels like freedom: without ε, the Goldstone boson is exactly massless, and lazy sliding costs nothing. The surprise

[3]The attentive reader might have also noticed too many oxford commas and certain weird comma placements — as if somebody replaced some other character.

quiz is a sudden, large ε. It is not a pedagogical tool. It is a mass generation mechanism. The author has experienced this mechanism firsthand and does not feel it was adequately disclosed at the time of admission.

6. Conclusion

We have shown:

- *(Representation theory)* The Lite Transformation acts via ρ_1 of $U(1)$. Its character traces S^1. You are going in circles. Your suffering lives in the trivial representation and is invariant under everything.
- *($\mathfrak{su}(5)$)* Life energy is conserved by unitarity. Taking it lite is the one-parameter subgroup e^{tH}, which decays CGPA as e^{-4t}. This is exact.
- *(Dynkin diagram)* CGPA and Social Life are the endpoints of A_4. Their direct connection would create a cycle, making the root system infinite-type. The Cartan–Killing classification rules this out. You cannot have both. This is not a lifestyle opinion. It is a theorem about Lie algebras.
- *(Spontaneous symmetry breaking)* The lazy state is the ground state of the Mexican hat potential. Procrastination is a Goldstone boson. The AUGSD is an explicit symmetry-breaking term. Surprise quizzes generate a mass gap. None of this was your fault. It was the vacuum.

The author would like to formally submit this paper as evidence in any future academic standing review. The degradation was not a personal failing. It was a consequence of the ground state structure of the theory, as proven above. The author is, in this sense, blameless. (The CGPA, however, remains.)

The mathematics does not judge you.

It simply computes.

It has computed a circle.

References

1. That one senior who said "bhai compre mein sab ho jaata hai" and then got a 4 CG. (2022). *Personal communication.* ANC, 2 AM.
2. The AUGSD timetable committee. (Every semester). *Timetable v1.0, v1.1, ..., v1.7.* Released progressively later with each revision until all your timetable plans are completely ruined. BITS Pilani.
3. The PS2 station allotment portal. (2024). *Internal error 500.* PSMS, BITS Pilani.
4. Student who attended every lecture, made color-coded notes, and got the same grade as the person in [1]. (2023). *Unpublished grievance.*
5. That one reference which the AI hallucinated. (2026) *Context Window.* Claude.